



Bridging the Awareness–Action Gap in Research, Innovation, and Entrepreneurship Competencies: Evidence from Saudi University Students in the Context of Vision 2030

Ayesha Siddiqha Mukthar^{1,2*}, Athari Abdulrahman Almuraikhi^{1,2}, Vinoth Raman³,
Wafa M. Al-Saleh^{1,2}

¹ King Abdullah International Medical Research Center, King Saud bin Abdulaziz University for Health Sciences, 31982 Al-Ahsa, Saudi Arabia

² Ministry of National Guards-Health Affairs, 31982 Al-Ahsa, Saudi Arabia

³ Deanship of Quality and Academic Accreditation, Imam Abdulrahman Bin Faisal University, 31441 Dammam, Saudi Arabia

* Correspondence: Ayesha Siddiqha Mukthar (mukthara@ksau-hs.edu.sa)

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Abstract: The development of research, innovation, and entrepreneurship (RIE) competencies has been positioned as a strategic priority within Saudi Arabia’s Vision 2030; however, a persistent discrepancy between awareness and active engagement remains insufficiently characterised. In this study, the levels of RIE awareness, perceptions, and experiential participation among university students in Saudi Arabia, with particular reference to the Eastern region, were systematically examined, and their statistical associations with competency development were evaluated. A cross-sectional survey design was employed, in which data were collected from 301 students during April–May 2025 using a validated 24-item, five-point Likert-scale instrument encompassing five constructs: RIE awareness, influencing factors, perceptions and attitudes, educational experiences, and sustainability orientation. High internal consistency was demonstrated (Cronbach’s $\alpha = 0.89$ – 0.93), and construct validity was assessed through exploratory factor analysis (EFA). Descriptive statistics indicated that RIE awareness was moderately high ($M = 3.54$, $SD = 1.00$), whereas a pronounced participation gap was observed: although 56.6% of respondents reported involvement in research activities, substantially lower engagement was recorded in innovation and entrepreneurship initiatives (24.9%) and start-up activities (19.2%). Perceived importance of RIE for future career development was high ($M = 4.13$), yet awareness of entrepreneurial mindset constructs remained comparatively limited ($M = 3.15$). Significant positive correlations were identified among the principal constructs (Spearman’s $\rho = 0.666$ – 0.902 , $p < 0.001$), although potential inflation effects attributable to shared measurement items were noted and critically considered. Ordinal logistic regression analysis revealed that participation in research projects and exposure to structured educational experiences constituted the most robust predictors of RIE competency development, surpassing attitudinal variables in explanatory power. These findings suggest that favourable perceptions alone are insufficient to foster competency acquisition in the absence of sustained experiential engagement. It is therefore implied that higher education institutions should prioritise the integration of practice-oriented RIE programmes, strengthen mentorship quality, and enhance transparency in resource accessibility, with policy interventions oriented towards capability development rather than motivational reinforcement. The study provides an empirically grounded baseline for assessing RIE competencies in emerging higher education contexts and offers a transferable measurement framework applicable to Gulf and comparable innovation-driven economies.

Keywords: Research competencies; Innovation capabilities; Entrepreneurship education; Saudi Vision 2030; Higher education systems; Experiential learning; Self-efficacy; Institutional support

JEL Classification: I23, L26, M13, O35

1. Introduction

The 21st-century knowledge-based economy requires a paradigm shift in higher education to provide students not just with academic knowledge, but also with the research, innovation, and entrepreneurship (RIE) skills needed to become productive members of national innovation ecosystems (Ratten & Usmanij, 2021). In Saudi Arabia, the national development agenda, Vision 2030, identifies knowledge-based industries and entrepreneurship as priority drivers of economic diversification, and universities as critical institutional players in the national innovation system (Alshumaimri et al., 2010). This national project demands a shift from theoretical to applied skills in RIE development, aligned with global Sustainable Development Goals (SDGs) and national transformation goals (Leal Filho et al., 2018).

Despite substantial investments in entrepreneurship centres, industry linkages, and start-up funding schemes, policy commitments to RIE skill development have not been effectively translated into classroom practice within Saudi universities (Ministry of Education, 2021). A structural gap has been identified between entrepreneurship education research, which predominantly emphasises pedagogical approaches, and the entrepreneurial university model, which prioritises governance structures and resource systems, thereby constraining the quality of RIE learning (Guerrero et al., 2016). Furthermore, RIE capability development has been hindered by academic reluctance to adopt active learning approaches, as well as by the continued reliance on outdated theoretical frameworks (Alhubaishy & Aljuhani, 2021).

This study addresses this gap by treating institutional support as an innovation capacity mechanism, not merely an educational convenience. Universities that invest in mentorship infrastructure, resource visibility, and structured experiential learning pathways function as innovation intermediaries that can convert student awareness into entrepreneurial action. The empirical question addressed in this study is where and how this mechanism breaks down: awareness, attitudes, skills, or institutional support.

The study examines: (1) student RIE awareness and self-assessment of skills; (2) factors that influence student RIE development; (3) the impact of attitudes/perception on learning; (4) the role of educational experiences in entrepreneurial intentions; and (5) student perceptions of RIE skills aligning with Vision 2030 and SDG goals.

Research, Innovation, and Entrepreneurship Awareness and Knowledge Foundation

Student participation in RIE requires RIE awareness (Zhao et al., 2022). Research skills include data analysis, critical thinking and academic writing; innovation skills include problem-solving, ideation and prototype development; entrepreneurial skills include strategic planning, market analysis and risk management (Bagaka's et al., 2015). Alhubaishy & Aljuhani (2021) label a positive attitude toward innovation as a prerequisite to the conditions that allow innovative ideas to flourish. RIE awareness is also beneficial for professional skill development: research engagement offers critical thinking, project management and communication skills relevant to various job roles (Ceyhan & Tillotson, 2020). Understanding of sustainability aspects also equips students for leadership in responsible innovation (Hsu & Pivec, 2021).

Entrepreneurial Intentions and Self-Efficacy

Intentions to start a new business (entrepreneurial intentions) are known precursors of entrepreneurial activity (Krueger et al., 2000). Self-efficacy (the belief in one's abilities to perform business-related activities) is an intermediary variable between education and entrepreneurial intentions (Neneh, 2022). Vision 2030 incentive structures amplify entrepreneurial intent among Saudi students, but sustained motivation requires both institutional support and practical skill development to convert intention into action (Almobaireek & Manolova, 2012). Research consistently shows that personal efficacy beliefs are central to persistence in the face of entrepreneurial challenges (Wu et al., 2022), and that mentorship, role-based learning, and simulation are effective self-efficacy-building strategies in Saudi institutional contexts (Islam & Khan, 2024).

Educational Experiences and Experiential Learning

Experiential learning theory (Kolb, 2014) establishes that competency development requires the active integration of theory and practice through structured reflection. Saudi institutions have increasingly prioritized practical learning environments to strengthen graduate employability and entrepreneurial capacity (Islam & Ali Khan, 2023). Training programmes, including seminars, business simulations, and incubator access, have demonstrated effectiveness in building entrepreneurial objectives and skills (Santana-Domínguez et al., 2022). However, technology adoption skills remain deficient among both students and faculty (Alhubaishy & Aljuhani, 2021), and mentoring remains limited in Saudi universities (Ceyhan & Tillotson, 2020).

Institutional Support as an Innovation Mechanism

Universities serve not just as educational institutions but also as part of national innovation ecosystems, offering resources, mentoring and information networks that help students develop research, innovation and entrepreneurship (Guerrero et al., 2016). Innovation infrastructure—access to financing, lab spaces, mentorship and entrepreneurship centre programs—directly affects the ability of students to translate ideas into a business. Insufficient institutional infrastructure, even with strong intentions, represents a system-level innovation capacity constraint rather than an educational shortcoming (Alhubaishy & Aljuhani, 2021).

Recent research on university innovation ecosystems (Audretsch & Belitski, 2020) highlights that institutional

support is dependent on systems integration: entrepreneurship centres, research offices, industry engagement, and more must all work in concert rather than in silos. Saudi universities have made substantial investments in entrepreneurship centre infrastructure under Vision 2030, yet the gap between research participation and startup activity documented in this study suggests that support systems remain disconnected from innovation translation pathways (Islam & Khan, 2024). Bagaka's et al. (2015) further emphasize faculty mentorship and structured research experiences as essential system-level supports, particularly for early-career students.

Attitudes, Perceptions, and Innovation Culture

Beliefs and attitudes are critical moderators of educational and entrepreneurial successes (Ajzen, 1991). Favorable attitudes towards innovation encourage creativity, risk-taking and decision-making (Blenker et al., 2011). Although attitudes to innovation in Saudi Arabia are shown to be changing, conservative attitudes may still limit entrepreneurial risk-taking in some cases; mentorship and role models are effective in modifying these attitudes (Almobaireek & Manolova, 2012). Engagement in research and mentorship improves learning and research performance, with research cultures encouraging innovation (Bagaka's et al., 2015). Cross-disciplinary research cultures, with appropriate infrastructure, also promote innovation (Ceyhan & Tillotson, 2020).

Technology Readiness and Digital Transformation

Digital readiness—the willingness to use digital platforms—affects the extent to which RIE tools and platforms are used by students (Purwati et al., 2022). Digital competency enables more effective research execution and innovation implementation (Oli, 2023). Despite significant investment in digital infrastructure under Saudi Arabia's Vision 2030, the digital divide needs to be addressed through equalization efforts, such as smart campuses (Islam & Khan, 2024). The embedding of digital entrepreneurship education—from virtual incubators to data analytics and online mentoring—is an emerging field with considerable potential for scalable RIE skill training (Islam & Ali Khan, 2023). Technology readiness is thus a student attribute and a systemic factor for RIE.

Conceptual Framework and Research Gaps

Five theoretically supported constructs—RIE awareness, key development factors (institutional support and resources), attitudes and perceptions, educational experiences, and sustainability alignment interact in the development of RIE competency as an associative model appropriate for a cross-sectional survey. Vision 2030 and the SDGs provide the national policy backdrop that gives these constructs their relevance in the Saudi context.

The model fills a gap in the literature: while the components (self-efficacy, experiential learning, institutional support, innovation culture, technology readiness) are well documented in the literature and in isolation, their composite relationship in the Gulf higher education environment—particularly in a time of rapid policy-based change - has not been empirically investigated. The present research addresses this gap in the literature through a baseline assessment that can guide intervention design.

2. Research Questions

RQ1: What are the levels of RIE awareness and self-assessed knowledge among Saudi university students in the Eastern region?

RQ2: Which institutional and personal factors are perceived as most important for RIE development?

RQ3: How do attitudes and perceptions towards RIE relate to self-reported learning outcomes?

RQ4: How do prior educational experiences relate to entrepreneurial intentions and RIE competency development?

RQ5: How do students perceive the alignment of their RIE competencies with Saudi Vision 2030 objectives and the SDGs?

3. Methodology

3.1 Research Design and Sampling

We conducted a cross-sectional survey in Saudi universities in April–May 2025 (30 days of data collection). Participants were recruited using convenience sampling across all levels and fields of study. Participants are heavily represented from the Eastern region (84.7%) and are mostly females (94.7%), which precludes generalization to all university students in Saudi Arabia; results should be viewed as pertinent to this sub-population. The study was approved by the Institutional Review Board, King Abdullah International Medical Research Center (KAIMRC), Protocol NRA25/011/2, IRB No. 00000134725.

3.2 Instrument Development

The questionnaire had three parts: (1) demographics (13 items); (2) history of participation in RIE; and (3) 24 statements rated on a 5-point Likert scale (1 = Strongly Disagree to 5 = Strongly Agree) in five categories. Qualitative feedback was sought via three open questions. The questionnaire was developed in English and

translated into Arabic by a language specialist with a bilingual background, with back-translation to ensure equivalence. Two RIE education experts evaluated content validity. Piloting ensured clarity, ordering and response consistency. Data were collected via Google Forms.

Table 1 describes the instrument, including the item sources and adaptations. Three of the constructs adapted items from a previously validated scale: Research self-efficacy items were adapted from Ceyhan & Tillotson (2020); Entrepreneurial Intentions items were adapted from Krueger et al. (2000); Innovation Attitudes items were adapted from Blenker et al. (2011). Two constructs—key factors and sustainability alignment—were created for this study, based on the Vision 2030 policy and reviewed by two RIE education experts. All adaptations were minor and were tested in the pilot test.

Table 1. Instrument description: constructs, sources, and adaptation status

Construct	Items	Original Source	Adaptation Status	Justification
Research, Innovation, and Entrepreneurship (RIE) Awareness	6	Bagaka's et al. (2015); Zhao et al. (2022)	Adapted	Items contextualised for Saudi higher education setting; two new items added for Vision 2030 alignment
Key Factors	6	Developed for this study	Newly developed	Based on Vision 2030 policy framework; reviewed by two RIE domain experts
Attitudes/Perceptions	4	Blenker et al. (2011); Ajzen (1991)	Adapted	Theory of Planned Behaviour items contextualised for innovation/entrepreneurship in higher education
Educational Experiences	4	Ceyhan & Tillotson (2020); Kolb (2014)	Adapted	Experiential learning items adapted for Saudi research and innovation programme contexts
Sustainability Alignment	4	Developed for this study	Newly developed	Items reflect Sustainable Development Goals (SDGs) targets 4, 8, and 9; reviewed by two sustainability education specialists

3.3 Data Analysis

Data was analyzed using IBM SPSS Version 31. Cases with missing data were removed from analyses. Data were described using frequencies, means, standard deviations, medians and IQR. Cronbach's alpha assessed internal consistency. Inter-construct associations were assessed by Spearman's correlation, because the data were ordinal.

The overall RIE development score was calculated as the mean of the 24 Likert-scale items from the five constructs. Given that this overall score contains items from each of the five sub-constructs, the correlations between sub-constructs and the overall score are, in part, artifactual and should not be interpreted as independent results.

We also used an exploratory factor analysis (EFA) based on principal axis factoring with varimax rotation to investigate the construct structure. A five-factor solution accounted for 68.3% variance. Items loaded most heavily on their assigned factors (loadings ≥ 0.45) and cross-loaded at less than 0.30, suggesting that the constructs are conceptually distinct. See Appendix C for the full EFA results.

Ordinal logistic regression was used to model the predictors of overall RIE development (categorized as low ≤ 2.99 ; moderate 3.00–3.99; high ≥ 4.00). This trichotomy was based on the theoretical and empirical middle of the 5-point scale. The predictors were previous research project involvement, innovation/entrepreneurship programme involvement and business/startup activity, with demographic controls (year of study, level of study, region) to consider the convenience sampling issue. We used an alpha level of $p < 0.05$ (95% confidence interval (CI)).

4. Results

4.1 Demographic Profile

Sample characteristics ($N = 301$) are shown in Table 2. The majority of participants were aged 18–20 years (73.7%) and female (94.7%). The majority were studying in universities in the Eastern region (84.7%), followed by the Central region (9.3%) and other regions (6.0%). Year of study: first year 35.2%, second year 32.4%, third year or higher 32.4%. Programme type: bachelor's degree 56.9%, paraprofessional degrees 39.5%. GPA: Excellent (4.75/5.00 and above) 23.1%, Superior (4.50/5.00 and above) 21.4%.

For prior RIE experience, 56.6% had participated in research projects, 24.9% in innovation/entrepreneurship programmes, and 19.2% in business/startup activities. Kruskal-Wallis tests showed statistically significant differences in overall RIE development by participation in research projects ($p = 0.007$) and innovation/entrepreneurship programmes ($p = 0.010$), but not startup activity ($p = 0.304$). These findings suggest

that students who have participated in research projects and innovation/entrepreneurship programmes report greater overall RIE development; the association with startup activity is unclear, perhaps due to the small number of students in this subgroup.

Table 2. Demographic characteristics and prior research, innovation, and entrepreneurship (RIE) participation (N = 301)

Items	Characteristic	Frequency	%	Median (Interquartile Range)
Age	18–20	207	73.7	3.75 (1.30)
	21–23	67	23.8	3.68 (1.17)
	24–26	6	2.1	3.45 (1.33)
Gender	Female	266	94.7	3.73 (1.28)
	Male	15	5.3	3.75 (1.40)
	Eastern	238	84.7	4.02 (0.83)
Region	Central	26	9.3	3.75 (1.34)
	Other	17	6.0	3.73 (1.59)
Prior Research Participation	Yes	159	56.6	3.56 (1.30)
	No	122	43.4	3.89 (1.02)
Innovation/Entrepreneurship Programs	Yes	70	24.9	4.03 (1.13)
	No	211	75.1	3.65 (1.27)
Business/Startup Activities	Yes	54	19.2	3.88 (1.15)
	No	227	80.8	3.68 (1.30)

4.2 Research, Innovation, and Entrepreneurship Awareness and Attitudes (Item-Level Results)

A descriptive analysis of individual items showed interesting trends. For recognition of importance, 47.7% strongly agreed RIE skills are important for personal and professional growth. Regarding university support, 41.6% strongly agreed that their university offers extracurricular activities to help develop RIE. Regarding career relevance, 35.6% strongly agreed that entrepreneurship is important in career development. Regarding future relevance, 31.3% strongly agreed current activities would fulfil Vision 2030 goals. Regarding social impact, 49.8% strongly agreed entrepreneurial activities have a positive impact on social and environmental issues.

4.3 Construct Reliability and Descriptive Statistics

Descriptive statistics and indices of reliability for the five constructs are shown in Table 3. The internal consistency of the constructs was high (Cronbach's $\alpha = 0.89$ – 0.93), showing that the scales were reliable. Average ratings were moderate (3.37–3.63) with large within-sample variability (Standard Deviation = 1.03–1.15). This would suggest that students are not strongly averse to nor strongly favorable of RIE engagement, which is consistent with a population that is just beginning their exposure to RIE, rather than a population that has barriers to accepting RIE.

Table 3. Descriptive statistics and reliability for research, innovation, and entrepreneurship (RIE) constructs

Construct	Items	Mean	Median	Standard Deviation	Cronbach's α	Exploratory Factor Analysis Variance %
Awareness	6	3.37	3.50	1.07	0.91	12.4%
Key Factors	6	3.53	3.67	1.03	0.89	13.1%
Attitudes/Perceptions	4	3.60	4.00	1.14	0.93	14.7%
Educational Experiences	4	3.58	3.75	1.14	0.92	13.8%
Sustainability Alignment	4	3.63	4.00	1.15	0.90	14.3%
Overall RIE Development	24	3.54	3.73	1.00	0.93	-

4.4 Highest and Lowest Rated Items

The top- and bottom-rated survey items are listed in Table 4. All of the highest-rated items reflect motivation and values ($M \geq 3.89$); all of the lowest-rated items reflect specific skill awareness and process knowledge ($M \leq 3.37$). The average score for the highest motivational item ($M = 4.13$, entrepreneurial social impact) is 0.98 scale points higher than the lowest skill-awareness item ($M = 3.15$, entrepreneurial mindset knowledge). This combination of strong motivation and lower skill awareness represents the main empirical contribution of the study and will be discussed.

Table 4. Top and bottom rated survey items

Top 5 Highest Rated Items	Mean	Interpretation
I believe entrepreneurial initiatives can create a positive social and environmental impact.	4.13	Strong value alignment; social entrepreneurship orientation
I believe research and innovation are crucial for personal and professional growth.	4.04	High intrinsic valuation of research, innovation, and entrepreneurship (RIE)
Practical learning opportunities... enhance my entrepreneurial intentions.	4.00	Demand for experiential engagement
Research and innovation can help achieve the Sustainable Development Goals (SDGs).	3.98	Perceived alignment with global/national goals
I perceive entrepreneurship as an important skill for achieving career success.	3.89	Positive career relevance perception
Bottom 5 Lowest Rated Items	Mean	Gap/Challenge
I am aware of the skills associated with ‘entrepreneurial mindset’.	3.15	Significant knowledge gap—lowest-rated item
I am committed to contributing to sustainability through research and innovation.	3.22	Action-belief disconnect
My personal interest in research, innovation, or entrepreneurship has helped my development in these areas.	3.28	Need for structured programmes
I am familiar with the process of conducting research and developing entrepreneurial projects.	3.35	Process knowledge deficit
I am aware of the skills of ‘innovation’.	3.37	Core skill gap

4.5 Correlation Analysis

Table 5 shows the matrix of Spearman’s correlations for each of the five constructs and the overall RIE composite. Inter-construct correlations were positive and significant ($\rho = 0.666$ – 0.805 , $p < 0.001$). Educational Experiences-Attitudes ($\rho = 0.805$), Key Factors-Attitudes ($\rho = 0.783$), and Key Factors-Educational Experiences ($\rho = 0.784$) were the strongest inter-construct correlations. These findings indicate that students who perceive enriching educational experiences also perceive greater positive attitudes towards innovation and greater perceptions of the importance of key factors—a consistent finding with experiential learning (Kolb, 2014).

Correlation between each sub-construct and the overall RIE development composite ($\rho = 0.858$ – 0.902) are high, as they should be since the composite is calculated from the same items as the sub-constructs (see Section 3.3). These verify instrument internal consistency but must not be interpreted as substantive results.

Table 5. Spearman’s correlation matrix for research, innovation, and entrepreneurship (RIE) constructs

Construct	1	2	3	4	5	Overall
1. Awareness	-					0.858**
2. Key Factors	0.716**	-				0.892**
3. Attitudes	0.695**	0.783**	-			0.899**
4. Educational Experiences	0.666**	0.784**	0.805**	-		0.902**
5. Sustainability Alignment	0.714**	0.750**	0.783**	0.796**	-	0.897**

Note: ** $p < 0.001$ (2-tailed).

4.6 Regression Analysis

Table 6 shows the ordinal logistic regression. The dependent variable was the overall RIE development, which was trichotomised into low (≤ 2.99), moderate (3.00–3.99), and high (≥ 4.00) based on answers at the below-middle, middle, and above-middle scores on the 5-point scale. This grouping of cells provides meaningful levels of performance and a sufficient number of cells. All predictors will use the reference category of the No participation one. $\text{Exp}(B) = 1.0$ means greater odds of higher category of RIE development; $\text{Exp}(B) = 0.0$ means lesser odds.

The participation in research projects was a statistically significant positive predictor of the development of RIE. Former research students were found to be more likely to report neutral or higher levels of development in RIE ($\text{Exp}(B) = 2.546$ – 3.047 , $p = 0.004$ – 0.016). This is the most predictive and consistent in the model.

The positive but insignificant effects were those on innovation/entrepreneurship programme participation ($\text{Exp}(B) = 1.864$ – 2.083 , $p = 0.095$ – 0.152). This direction is as expected, and the trend could be due to a lack of statistical power with the use of the small subgroup ($n = 70$, 24.9% of the sample).

There was an ambivalent trend in business/startup activity. Students who participated in start-up activity were less likely to report strongly disagree/disagree vs. strongly agree/agree ($\text{Exp}(B) = 0.256$, 95% CI [0.109, 0.602], $p = 0.002$), and more likely to report neutral vs. strongly agree/agree ($\text{Exp}(B) = 3.047$, 95% CI [1.491, 6.227], $p = 0.002$). The probable reason is that students who engage in a startup activity without proper preparation can feel

less competent in their own judgment, but, again, this interpretation was not directly examined in this experiment.

The demographic controls, year of study (3rd year and above versus 1st–2nd year) was a marginally significant positive predictor ($\text{Exp}(B) = 1.67$, $p = 0.048$), which indicated that the development of the RIE is positively related to the academic progression. This model did not show statistical significance in programme level and region.

The overall model fit was high ($\chi^2 = 31.45$, $p = 0.001$), which proves that the interaction of the participation history and demographic factors explains the significant variance of the results of the RIE development.

Table 6. Ordinal logistic regression predicting overall research, innovation, and entrepreneurship (RIE) development

Predictor Variable	Reference Category	Exp (B)	95% Confidence Interval	p-Value
Research Project Participation (No vs. Yes)	SD/D vs. SA/A	0.256	[0.109, 0.602]	0.002**
	Neutral vs. SA/A	2.546	[1.193, 5.433]	0.016*
Innovation/Entrepreneurship Programs (No vs. Yes)	SD/D vs. SA/A	1.864	[0.796, 4.365]	0.152
	Neutral vs. SA/A	2.083	[0.880, 4.929]	0.095
Business/Startup Activities (No vs. Yes)	SD/D vs. SA/A	0.770	[0.290, 2.047]	0.601
	Neutral vs. SA/A	3.047	[1.491, 6.227]	0.002**
Year of Study (1 st –2 nd vs. 3 rd +) [Control]	Neutral vs. SA/A	1.670	[1.005, 2.775]	0.044*
Programme Level (Bachelor's vs. PG) [Control]	Neutral vs. SA/A	1.234	[0.712, 2.139]	0.0453
Region (Eastern vs. Other) [Control]	Neutral vs. SA/A	1.115	[0.589, 2.110]	0.737

Note: SD/D = Strongly Disagree/Disagree; SA/A = Strongly Agree/Agree; Model $\chi^2(6) = 24.73$, $p = 0.001$.

4.7 Open-Ended Responses (Supplementary Qualitative Evidence)

Three open-ended questions were added in order to describe subtle perceptions that are not captured by Likert-scale items. They were analyzed through a two-step thematic analysis of responses ($N = 301$): (1) open coding of the individual responses into descriptive categories; (2) thematic sorting of codes into higher-order groups. A 20% random sample was coded by two researchers who were independent; the inter-rater reliability was estimated at $\kappa = 0.81$ (substantial agreement). The themes that were developed:

Theme 1: RIE Impact on Academic Performance

The majority of answers (22.3%, $n = 67$). The participants explained that engagement in RIE resulted in positive academic outcomes, such as enhanced critical thinking, acquisition of knowledge, and motivation. One typical answer: “The experience of working on a research project taught me to be more cautious in information analysis and think in a more independent way than any course did.” Another 16.3% ($n = 49$) mentioned the benefits of creativity and problem-solving. It is important to note that 30.2% ($n = 91$) said they were not affected by it, and most of them did not attend RIE activities.

Theme 2: Challenges and Barriers

Barriers that were reported were time management barriers (1.3%, $n = 4$), lack of workshops and training opportunities (0.3%, $n = 1$), and personal motivation (1.3%, $n = 4$). The non-response rate of this question (29.2%, $n = 88$) can be high because of lack of awareness of certain barriers or an overall impression of not having worked enough to be exposed to the barriers. An example of a quote: “I would like to be engaged in the field of innovation, but I even do not know where to begin and whom to address”.

Theme 3: Recommendations

The most common suggestion was institutional assistance (27.6%, $n = 83$), which includes practical training, mentorship, access to laboratories, funding and industry networks. One of the typical answers: ‘The university should assign a mentor to each student interested in research or entrepreneurship, not just make the resources available on a website’. Other suggestions were awareness campaigns (17.3%, $n = 52$), personal initiative encouragement (8.0%, $n = 24$), and industry-education partnership (2.3%, $n = 7$). The non-response rate to this question was 33.6% ($n = 101$).

5. Discussion

5.1 The Awareness–Action Paradox: The Central Finding

The most significant result of this study: Saudi university students in the Eastern region express unconditionally positive views on RIE and attribute significant importance to its benefits, but in fact, few of them have a significant experience of engaging in the sphere of innovation or a startup. Entrepreneurial social impact is perceived to be highly rated by students ($M = 4.13$), and RIE is seen as a crucial element in personal and professional development

($M = 4.04$), but the level of awareness of entrepreneurial mindset is lower ($M = 3.15$), and less than a fifth of students (19.2%) report having participated in a startup. This difference of 0.98 between the highest motivational item and the lowest skill-awareness item is not a fringe discrepancy, but rather a structural mark of an education system that has already done the communicating of the worth of RIE without yet having to bring the structural scaffolding where attitudes transform into competencies.

Interest in the participation gradient, namely, the research project (56.6%) to the innovation programmes (24.9%) to the startup activities (19.2%). It implies that passive academic exposure (research incurred in the curriculum) does not necessarily lead to proactive entrepreneurial participation. This observation is in line with Wu et al. (2022), who indicated that entrepreneurship education does not have a direct influence but mediates self-efficacy to generate behaviour. The regression analysis supports this relationship: prior research participation has the greatest and positive impact on increased development of RIE ($\text{Exp}(B) = 2.546\text{--}3.047$, $p < 0.016$), compared to innovation programme participation, which has a positive but insignificant effect, which may be due to lack of power of the subgroup.

The correlation between the negative relationship between the startup activity and some comparison of RIE development should be considered carefully. The most likely reason is that students whose start-up activity lacks proper preparation feel that they are any less competent than they used to be - a ceiling effect of attempted practice on the lack of underlying training. This interpretation is a speculation and not directly measured in this data; the next wave of longitudinal study can focus on testing whether the engagement in startups can predict the development of competence with time after adjusting the level of preparation.

5.2 Institutional Support as an Innovation Capacity Constraint

The institution-level beliefs held by students were moderate to positive: 41.6% strongly believed their university offers extracurricular RIE activities, and 32.4% indicated a lack of sufficient resources. The importance of financial incentives was supported by 31.7% of students, which should be compared with Li et al. (2020), who also found funding to be a strong determinant in the field of entrepreneurship. The institutional conditions are central because of the significant relationship between the Key Factors construct and the overall development of the RIE ($\rho = 0.892$, $p < 0.001$).

Nevertheless, with the large number of neutral answers to most institutional support items, it appears that the existence of support is not communicated well to students, or that its coverage is not yet at the level of innovation and venture creation routes. That, in terms of innovation systems (Guerrero et al., 2016), is a capacity constraint: higher education institutions might have the structures required, but without their ability to access, navigate, and utilize them easily, the innovation system itself is a closed system to a large number of potential actors. A similar point is made by Islam & Khan (2024) when it comes to the digital infrastructure growth in Saudi entrepreneurship centers: resource availability and the lack of accessibility are not effective examples of institutional support.

Mentorship was rated neutrally to moderately, with a middle-range focus. One reason could be that faculty mentors are only available in research-oriented activities and have never been trained or incentivized to assist in the innovation-to-venture transition. The other reason could be that the students do not know of possible mentorship channels available. Both options, once proved by future studies, would point the blame at various intervention strategies: mentor training and student outreach, respectively.

5.3 Educational Experiences and the Experiential Learning Gap

Attitudes were most closely related to the educational experiences ($\rho = 0.805$), and attitudes were reported to have a connection with overall RIE development in line with the experiential learning theory (Kolb, 2014). Students greatly supported practical learning experiences ($M = 4.00$) and supported that exposure to a real-world problem develops entrepreneurial abilities (32.4% strongly agree), but only 24.9% had undertaken innovation programmes. It is this disjuncture between endorsement and participation that is the experiential learning deficit: students appreciate access to hands-on education but are not being accorded enough structured access to do so.

The medium value of the contribution of academic courses to the development of RIE ($M = 3.53$) indicates that formal curriculum, although to some extent useful, is oriented more to theoretical than to applied learning. This is supported by the open-ended questions: 27.6% of students mention practical training, workshops, and mentorship as the main recommendation, meaning that these students do not see the curriculum as a means of competency development. The result is similar to Santana-Domínguez et al. (2022) about the effectiveness of structured training programmes and Alhubaishy & Aljuhani (2021) about the lack of technology adoption competencies in Saudi higher education.

5.4 Perceived Alignment with Vision 2030 and the Sustainable Development Goals

Students rated their perceived consistency with the targets of Vision 2030 (31.3% strongly agree, $M = 3.88$) and

the role of RIE in contributing to the SDGs achievement (39.1% strongly agree, $M = 3.98$) as being high. The 49.8% who strongly agree that entrepreneurial initiatives have a positive impact on social and environmental outcomes ($M = 4.13$) indicates that the values of sustainability are significantly internalized, which is unsurprising in accordance with Hsu & Pivec (2021) about sustainability orientation and entrepreneurship education.

The sustainability alignment result does, though, point out the 0.91-point difference between supporting sustainability impact ($M = 4.13$) and individual commitment to donating through RIE ($M = 3.22$). It is the biggest item-level gap of the research. The students seem to support sustainability objectives on a high level and do not feel sure about their individual potential of making a contribution using RIE. This aligns with Neneh (2022) about the contribution of self-efficacy to the translation of entrepreneurial values into intentions: the absence of practical competence does not transform the approval of national goals into entrepreneurial action.

5.5 Technology Readiness as a System-Level Enabler

Technology readiness was not directly assessed as a key construct in this study; nevertheless, the enabling conditions in the system level, identified through open-ended responses and the literature, are technology readiness. Improved digital infrastructure in particular was specifically suggested by seventeen percent of students as a response to the open-ended question. Alhubaishy & Aljuhani (2021) recognized the lack of digital resources as one of the key barriers to integrating innovation in Saudi universities, and Purwati et al. (2022) and Oli (2023) report the importance of technology preparedness in the adoption of RIE tools.

In terms of innovation systems, digital infrastructure is not only a student competency problem but a state of platform: student-centered universities that invest in smart campus technologies, virtual incubation platforms, and digital mentorship channels take their innovation systems outside their physical boundaries. The ongoing smart campus investments as part of the Vision 2030 provide a chance to reduce inequities in access, although equal access to these resources, especially for students who do not live in the Eastern states or have limited access to personal devices, must be designed. Technology readiness should be used as an acquired construct in future research in studies on RIE competencies.

5.6 Gender Considerations and Sample Characteristics

A sample of 94.7% predominantly female is representative of the gender proportion in the sampled universities and this is the trait of the Eastern region's higher education setting and not an aberration. The male subsample ($n = 15$) is too low to allow reliable gender comparisons; exploratory tests indicated that male students have slightly higher research self-efficacy, but the result should not be deemed a strong gender difference. According to Shinnar et al. (2014), gender differences in entrepreneurial self-efficacy are recorded across contexts, and cultural specifics in Saudi Arabia—which again is changing as part of the gender inclusion programs under Vision 2030—can moderate those trends. This question needs to be addressed through future research that would have a balanced gender sample.

The relative youth and low academic level of the sample (73.7% aged 18–20; 67.6% in years 1–2) also give significant background to the moderate levels of RIE awareness. Students that are in the early-career stage have not had much experience with the RIE competency-building curricula, programmes and mentorship structures. Longitudinal studies that would follow the development of RIE over academic years would help in clarifying whether the attitudinal-participation gap reduces with the progress of the students.

5.7 Theoretical and Practical Implications

In theory, the gap in competencies in the process of entrepreneurial development of the study group in the context of the four-stage framework of entrepreneurial development is identified: positive attitudes come before and not during skill development. This is an extension of intention-behaviour models (Ajzen, 1991; Krueger et al., 2000) since they record an intermediate stage where students are motivationally oriented to entrepreneurial objectives but are lacking in the underlying competency to fulfil them. Attitudinal readiness is a sufficient but not necessary factor to entrepreneurial behaviour; the intervening factor is institutional scaffolding.

The inter-construct correlations are high ($\rho = 0.666$ – 0.805): the development of RIE is systematically coherent: any interventions into a particular construct have a high probability of causing a positive ripple effect on another one. Nevertheless, research participation and structured educational experiences are mentioned as the most leverage points of intervention in the regression outcomes. These are the main targets that should be considered in the design of the programme.

In practice, Saudi universities should: (1) increase the number of experiential learning opportunities, including hackathons, incubators and venture challenges; (2) create pre-startup foundational education in entrepreneurial thinking and innovation processes; (3) redesign mentorship programs to encompass the entire research-to-innovation-to-venture pipeline rather than just the research part; (4) increase resource visibility, such as making

sure students can easily find and access RIE support; and (5) create inclusive programmes that reach students of all years of study and regions, not just those who self-select into innovation centers.

5.8 Study Limitations

First, the cross-sectional design takes a snapshot in time and cannot determine causality or the developmental course. They require longitudinal research to monitor the development of the attitude-participation gap throughout a career in academia. Secondly, the convenience sample is extremely localized in the Eastern (84.7%) region and is mostly female (94.7%), restricting generalization. The results must be understood as those of this particular sub-population of Saudi students instead of Saudi university students in the country.

Third, self-report measures have social desirability. Objective behavioral observations—existence of startup formation/formation, results of innovations, record of publications—would offer complementary validity in future research. Fourth, although the internal consistency is high, evidence of construct validity is based on EFA; a confirmatory factor analysis with a larger, more representative sample would provide more solid evidence to the measurement model.

6. Conclusions

6.1 Core Empirical Finding

This study reports a similar level of awareness–action gap among Saudi higher education students in the Eastern region: students have very positive views towards RIE, they report a high perceived fit with Vision 2030 and SDG goals, but report a low level of innovation programmes and start-up activities. The central empirical contribution of the 0.98-point difference between motivational endorsement and skill awareness and the participation gradient—research (56.6%) to innovation programmes (24.9%) to startup activity (19.2%)—is established. The best statistical predictors of RIE competencies development are research project participation and educational experiences, and all five constructs are highly inter-correlated, which demonstrates coherence in the system of RIE development.

6.2 Practical Implications for Research, Innovation, and Entrepreneurship Education in Saudi Higher Education

Motivational alignment is already in place, so interventions need to focus on skill building and participation more than awareness campaigns. Universities need to invest in training in pre-startup entrepreneurial thinking, programmed mentorship along the research-to-venture continuum, and pragmatic opportunities like innovation challenges and incubator programmes. Resource visibility—being aware of what RIE support is and its availability—is a priority that has been recognized across quantitative and qualitative data sets as low cost and high impact.

6.3 Limitations and Future Directions

The sample, which is concentrated in the Eastern region and female-dominated, is a limitation to generalization nationally. The cross-sectional design does not allow causal inferencing. Future studies are advised to: (1) sample nationally across regions of Saudi and types of institutions; (2) use longitudinal designs to trace the development of RIE over academic careers; (3) use gender-balanced samples to investigate differences in competency development; and (4) use objective behavioral outcome measures to supplement self-report data. The measurement evidence base would also be reinforced by confirmatory factor analysis on a larger sample.

Author Contributions

Conceptualization, A.S. and A.M.; methodology, A.S. and A.M.; software, V.R.; validation, A.S., A.M., and V.R.; formal analysis, A.S. and V.R.; investigation, A.M.; resources, A.M.; data curation, V.R.; writing—original draft preparation, A.M.; writing—review and editing, A.M.; visualization, A.S.; supervision, A.S.; project administration, A.S. All authors have read and agreed to the published version of the manuscript.

Informed Consent Statement

I/We confirm that, where applicable, all research involving human participants has been conducted in compliance with (e.g., the Declaration of Helsinki, Ethical approval from relevant Institutional Review Boards (IRBs), or Ethics Committees) and approved by the appropriate ethics committee or institutional review board.

Informed consent was obtained from all participants, and their confidentiality and privacy were strictly protected.

An informed consent was obtained from King Abdullah International Medical Research Center with IRB approval No#00000134725.

The study received an approval from King Abdullah International Medical Research Center (KAIMRC) with the study No#NRA25/011/2.

Data Availability

The data supporting our research results are under privacy or ethical restrictions. The data are available from the corresponding author for researchers who meet the criteria for accessing confidential data.

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Conflicts of Interest

The authors declare no conflict of interest.

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